Divyanshu Malik

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Summary

AI/ML and Software Engineer with 4+ years of experience building production-grade generative AI, machine learning, and distributed software systems. Specializes in RAG, agentic AI, LLM applications, and AI infrastructure, with end-to-end experience across data ingestion, retrieval, model orchestration, backend services, evaluation, deployment, and observability. Proven experience delivering clinical AI, research intelligence, and real-time computer vision systems using Python, C++, TypeScript, Azure, AWS, PyTorch, Kubernetes, and modern LLM and retrieval frameworks.

Experience

Amgen Inc.

Mar. 2026 – Present

AI Software Engineer II

United States

- Contributed to a production multi-agent clinical intelligence platform using Azure OpenAI, OpenAI Agents SDK, Azure AI Search, and MCP, supporting 2,400+ internal users across protocol review, patient cohort discovery, site feasibility, and evidence analysis.
- Implemented agent workflows that decomposed clinical questions into retrieval, eligibility analysis, cohort reasoning, site feasibility, evidence review, and response generation, integrating governed tools for Azure AI Search, SQL cohort queries, PubMed, ClinicalTrials.gov, and internal clinical datasets.
- Developed components of clinical-data ingestion pipelines using Azure Data Factory, Azure Functions, ADLS Gen2, Databricks, PySpark, and Delta Lake; integrated Azure AI Document Intelligence to extract eligibility criteria, endpoints, visit schedules, adverse events, tables, and amendment details for versioned downstream analysis.
- Implemented hybrid RAG in Azure AI Search combining vector search, BM25, semantic ranking, metadata filtering, and reranking; generated embeddings with Azure OpenAI and constructed citation-backed context to ground agent responses in protocols, clinical evidence, literature, and trial-registry data.
- Developed FastAPI services for protocol analysis, cohort matching, site ranking, and evidence summarization; containerized workloads with Docker and deployed them on AKS with Azure API Management, scalable replicas, health checks, and controlled rollouts for concurrent clinical-AI workloads.
- Implemented cohort-analysis capabilities comparing protocol eligibility criteria against de-identified patient attributes, diagnoses, medications, laboratory values, and clinical history, producing explainable eligibility results with supporting evidence and exclusion reasons.
- Implemented site-feasibility capabilities using historical enrollment, dropout trends, investigator performance, geography, disease prevalence, and prior trial metrics to support candidate site ranking and evidence-backed recruitment planning.
- Integrated Entra ID, RBAC, Managed Identity, Key Vault, Private Endpoints, Microsoft Purview, OpenTelemetry, Application Insights, Azure Monitor, and Log Analytics for application security, governance, and observability; evaluated RAG quality using MLflow, RAGAS, Recall@K, Precision@K, MRR, NDCG, groundedness, citation coverage, and clinical-review feedback.

Apache ResilientDB (Incubating)

June. 2025 – Mar. 2026

AI Software Engineer

United States

- Developed Nexus, an AI research assistant for the Apache Incubating ResilientDB project, enabling students and researchers across multiple universities to investigate distributed databases, blockchain consensus, and systems literature through citation-grounded RAG, multi-document research, and contextual code generation; reduced time-to-information by approximately 70%.
- Designed an AWS-native, event-driven document intelligence pipeline using Amazon S3, SQS, and Textract to extract text, tables, and document structure from research papers, technical documents, and code artifacts; generated text-embedding-3-large embeddings and stored vectors, metadata, and document versions in Amazon OpenSearch Serverless, supporting incremental indexing, content deduplication, and data freshness.
- Implemented hybrid retrieval combining BM25 lexical search with cosine-similarity vector search over HNSW indexes; used GPT-4o-mini for query rewriting and lightweight classification, followed by metadata filtering and context selection to improve retrieval for terminology-heavy and semantically complex distributed-systems queries.
- Built the RAG generation workflow with GPT-4.1 for complex reasoning, constructing responses from retrieved evidence and enforcing citation grounding; evaluated retrieval using Hit Rate@K, Recall@K, Precision@K, Context Recall, Context Precision, MRR, and NDCG, and generation using faithfulness, groundedness, citation accuracy, answer relevance, and hallucination rate.
- Established automated evaluation with golden datasets, LLM-as-a-Judge, and human review to validate model, prompt, and retrieval changes; used evaluation results to tune query rewriting, retrieval configuration, prompt templates, and context construction, with regression gates before production releases.
- Built an event-driven backend using FastAPI, separating ingestion, retrieval, memory, and generation into containerized services deployed on Amazon EKS; implemented asynchronous processing, horizontal autoscaling, health checks, retries, circuit breakers, idempotent ingestion, dead-letter handling, and Redis caching to support concurrent research workloads and real-time streaming.
- Implemented shadow, canary, and blue-green deployments through CI/CD with automated testing and RAG regression checks; monitored latency, time-to-first-token, token usage, cost per query, error rates, retrieval failures, and index freshness using CloudWatch, while securing services with IAM roles, Secrets Manager, private networking, TLS, and least-privilege access.

- Developed the research interface using Next.js, React, Tailwind CSS, and Radix UI, providing multi-document conversations, PDF previews, source citations, persistent session context, real-time streaming, and an interactive code composer.

National Center for Interventional Biophotonic Technologies
Sept. 2022 – Jun. 2025

Software Engineer – 3D Reconstruction and Computer Vision — United States

- Spearheaded an NIH-funded project for real-time cancer tumor visualization and surgical assistance using computer vision, 3D reconstruction, Fluorescence Lifetime Imaging (FLIm), PyTorch, ONNX, and YOLOv8, contributing to the $6M NIH P41 EB032840 research initiative.
- Developed a data-centric YOLOv8 instance segmentation pipeline for fiber-based FLIm aiming-beam detection in Transoral Robotic Surgery (TORS), addressing uneven illumination, specular reflections, contrast variation, and label noise across 40 in-vivo surgical videos; achieved an 85% median detection rate and 78% mean detection rate.
- Improved aiming-beam robustness through confidence-based data pruning and label-quality analysis, increasing detection to 93% on a challenging patient case while reducing the miss rate to 7%; selected a confidence threshold $\kappa = 0.2$ that removed 9% of training samples while improving model performance.
- Optimized the clinical FLIm detection pipeline for real-time inference, achieving 24 FPS across the complete visualization pipeline during TORS procedures; YOLOv8-n achieved 29.37 FPS with a 74% mean detection rate, providing an effective accuracy–latency trade-off for intraoperative surgical guidance.
- Developed a live incremental 4D Gaussian Splatting framework for monocular endoscopic reconstruction, processing 80–100 frames per batch and continuously adapting to camera motion, tissue deformation, occlusions, and changing surgical viewpoints; reduced reconstruction time from 579s to 34–60s per video.
- Integrated frame-aligned intraoperative FLIm measurements into evolving 3D reconstructions using TAPNext for temporal point tracking, DepthAnything for depth priors, and depth-guided backprojection to place FLIm measurements as colored Gaussian primitives, preserving functional tissue information under deformation and viewpoint changes.
- Achieved 30.15 PSNR, 0.86 SSIM, and 0.15 LPIPS with the 300-iteration incremental model, outperforming the offline Endo4DGS baseline (28.36 PSNR, 0.83 SSIM, 0.20 LPIPS) while producing usable modality-augmented 3D reconstructions within 30–60s versus 10+ minutes of post-hoc training.
- Built an interactive React.js, Flask, and Socket.IO visualization platform with batch-level model monitoring, video playback, reconstruction-quality metrics, and interactive 3D Gaussian Splatting views, enabling surgeon-in-the-loop inspection, checkpoint selection, and real-time exploration of reconstructed anatomy and functional FLIm overlays.

Technical Skills

Languages: C++, Python, JavaScript/TypeScript, Java, Kotlin, SQL, Bash

AI / ML: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, OpenCV, Pandas, NumPy, XGBoost, Feature Engineering, Classification, Regression, Model Evaluation

Generative AI: GPT-4.1, GPT-4o, GPT-4o-mini, Claude, Gemini, Llama 3, Mistral, LLMs, Prompt Engineering, Function Calling, Tool Calling, RAG, Agentic RAG, Multi-Agent Orchestration, MCP, OpenAI Agents SDK, LangChain, LangGraph, LlamaIndex, OpenAI API, Human-in-the-Loop Workflows

Retrieval & RAG: Hybrid Search, Dense Retrieval, BM25, Vector Embeddings, Reranking, Metadata Filtering, Chunking, Query Rewriting, Citation Grounding, RAGAS, Knowledge Graphs

Vector Search: Amazon OpenSearch Serverless, Azure AI Search, Pinecone, Weaviate, FAISS, pgvector, HNSW, ANN

Cloud: AWS, Amazon Bedrock, S3, EC2, Lambda, Glue, SageMaker, Amazon Redshift, CloudWatch, CloudTrail, Azure, Azure OpenAI, Azure AI Search, Azure Document Intelligence, Azure Monitor, Application Insights, GCP Vertex AI, BigQuery

Data Engineering: Apache Spark, PySpark, Databricks, Delta Lake, Kafka, Airflow, ETL/ELT Pipelines, Data Lakes, Data Warehouses, Batch Processing, Stream Processing

Backend & Full Stack: FastAPI, Flask, Node.js, Express.js, Next.js, React.js, REST APIs, Tailwind CSS, Radix UI, Socket.IO, WebSockets

Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Redis, Elasticsearch, Snowflake, Amazon Redshift

MLOps & DevOps: Docker, Kubernetes, Amazon EKS, AKS, MLflow, Git, GitHub Actions, CI/CD, Terraform, Linux/Unix, Model Monitoring, Drift Detection

Evaluation & Observability: Precision, Recall, F1, ROC-AUC, Recall@K, Precision@K, Hit Rate@K, Context Precision, Context Recall, MRR, NDCG, RAGAS, Faithfulness, Groundedness, Citation Accuracy, Error Analysis, LLM-as-a-Judge, OpenTelemetry, Prometheus, Grafana

Security: IAM, RBAC, Azure Entra ID, AWS KMS, AWS Secrets Manager, Key Vault, PII Handling, Guardrails, Auditability, Human Review Controls

Education

University of California, Davis
California, USA

Master of Science in Computer Science

Delhi Technological University
Delhi, India

Bachelor of Technology in Information Technology